C2 Bonding, Structure and Properties – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Types of Bonding **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Ionic bonding occurs between				Metal and non-metal
2	Covalent bonding occurs between				Non-metals
3	Metallic bonding occurs in				Metals
4	Ionic bond				Electrostatic attraction between oppositely charged ions
5	Covalent bond				Shared pair of electrons
6	Metallic bonding				Positive ions in a sea of delocalised electrons
7	What happens to electrons in ionic bonding				Transferred
8	Structure of an ionic compound				Giant ionic lattice
9	Formula of sodium chloride				NaCl
10	Formula of magnesium oxide				MgO

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 2: States and Properties **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Three states of matter				Solid, liquid, gas
2	State symbols				(s), (l), (g), (aq)
3	Meaning of (aq)				Dissolved in water
4	Limitations of the particle model				Particles shown as solid spheres; no forces shown
5	Why ionic compounds have high melting points				Strong electrostatic forces in all directions
6	When ionic compounds conduct				When molten or dissolved
7	Why they conduct then				Ions free to move
8	Why simple molecular substances have low boiling points				Weak intermolecular forces
9	Do simple molecular substances conduct?				No
10	Polymer				Very large molecule of many repeating units

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 3: Giant Covalent Structures and Carbon **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Examples of giant covalent substances				Diamond, graphite, silicon dioxide
2	Bonds per carbon in diamond				4
3	Why diamond is very hard				Every carbon has 4 strong covalent bonds
4	Bonds per carbon in graphite				3
5	Structure of graphite				Layers of hexagonal rings
6	Why graphite is soft				Layers slide (weak forces between layers)
7	Why graphite conducts				Delocalised electrons
8	Graphene				Single layer of graphite
9	Fullerenes				Hollow carbon molecules, e.g. C_{60}
10	Carbon nanotubes				Cylindrical fullerenes

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 4: Metals and Alloys **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Why metals conduct electricity				Delocalised electrons
2	Why metals conduct heat				Delocalised electrons
3	Why metals have high melting points				Strong metallic bonds
4	Why pure metals are soft				Layers of atoms slide easily
5	Alloy				Mixture of a metal with other elements
6	Why alloys are harder				Different sized atoms distort the layers
7	Example of an alloy				Steel
8	Why polymers are solid at room temperature				Large molecules, strong intermolecular forces
9	Bonds within polymer molecules				Strong covalent bonds
10	Why diamond does not conduct				No delocalised electrons

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C2 Bonding, Structure and Properties – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 5: Nanoparticles Paper 1 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Size of nanoparticles				1–100 nm
2	1 nm in metres				1×10^{-9} m
3	Size of fine particles ($PM_{2.5}$)				100–2500 nm
4	Size of coarse particles (PM_{10})				2.5–10 µm
5	Why nanoparticles behave differently				Very high surface area to volume ratio
6	Effect of side length ÷ 10 on surface area : volume				× 10
7	Uses of nanoparticles				Medicine, electronics, cosmetics, sun creams, catalysts
8	Use of silver nanoparticles				Antibacterial, e.g. wound dressings
9	Why less nanoparticle catalyst is needed				Larger surface area
10	Risk of nanoparticles				Unknown effects on health

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.